

Hexadecimal & BCD — Practice · AS 9618 — ANSWERS (student copy) · first edition, the sheet set as homework

1. ① $2B = 2 \times 16 + 11 = 43$ · ② $200 = C8$ · ③ 45 in BCD = **0100 0101** · ④ $1011\ 1101 = BD$ — 1 mark per row.
2. Working: $A3C = 10 \times 256 + 3 \times 16 + 12$ (or via binary $1010\ 0011\ 1100$). Answer: **2620** — 1 mark working, 1 mark answer.
3. **74D9** — nibble by nibble; 2 marks (lose 1 per wrong nibble, minimum 0).
4. (a) **0011 0000 0111** — the middle **zero nibble must appear**; that's the trap. [1] (b) BCD nibbles encode single denary digits 0–9 only; 1010 is 10, which is not a single digit — patterns A–F are forbidden in BCD. [1]
5. BCD: driving digit displays (calculators, meters, lift indicators — one nibble per 7-segment digit), or exact-decimal storage of currency. Hex: any of MAC addresses / memory addresses & dumps / colour codes / error codes — described, not just named. 1 mark each.
6. BCD: **0101 1000** (5 then 8) · Binary: **0011 1010** ($32+16+8+2$). Different patterns — BCD transliterates the digits, binary encodes the value. 1 mark both patterns, 1 mark the difference stated.

Total: 15 marks. Q4(a) trap: dropping the zero nibble gives 0011 0111. Q1 row ① trips the "hex 11 = eleven" crowd — 2B keeps them honest.



Ice-Cream Challenge — judge's notes + model solution

Award for: both directions · any length · left-padding handled · no `int()` / `hex()` / `bin()` . Model:

```
BITS = ['0000', '0001', '0010', '0011', '0100', '0101', '0110', '0111',
        '1000', '1001', '1010', '1011', '1100', '1101', '1110', '1111']
HEX = '0123456789ABCDEF'

def bin_to_hex(s):
    while len(s) % 4 != 0:
        s = '0' + s                # pad the LEFT nibble
    out = ''
    for i in range(0, len(s), 4):
        out = out + HEX[BITS.index(s[i:i+4])]
    return out

def hex_to_bin(s):
    out = ''
    for ch in s.upper():
        out = out + BITS[HEX.index(ch)]
    return out
```

Verified: `bin_to_hex('11010110') → D6` · `hex_to_bin('2F') → 00101111` · round-trips clean. Sixteen if-statements instead of the table still eats. Same deal returns in U6: signed denary ↔ binary, once two's complement lands.

🕒 **The quiet one, unhinted.** Question 4(b) asks why 1010 can never appear as a BCD nibble — and 1010 is **exactly hexadecimal A**, the very first hex digit that is not also a denary digit. The two halves of this sheet meet on the same four bits: the pattern where hex starts needing letters is the pattern where BCD runs out of digits. Same boundary, two notations, one bit pattern.